StudyRoom Connect

MikroTik Setup and Package Configuration Guide

This guide explains the improved package form, burst and schedule rules, MikroTik onboarding, remote Winbox access, provisioning commands, and day-to-day router management workflow.

Screenshot placeholder: MikroTik routers dashboard

1. System Overview

StudyRoom Connect manages ISP packages, customers, MikroTik devices, and provisioning commands from a single Laravel billing system.

The current implementation keeps the existing Blade ISP workflow while adding configuration fields and management actions that can later be connected to a RouterOS API worker.

- Packages define pricing, speed, burst, schedules, visibility, and router availability.
- MikroTik routers store provisioning status, remote Winbox details, service status, and last-seen information.
- Provisioning remains non-destructive and public fetch routes only return RouterOS script text.

Workflow: Package -> Router availability -> Customer provisioning -> RouterOS command

2. Package Creation

Create packages from Internet Packages. A package still requires the same core fields: name, price, speed, billing cycle, validity, and status.

The new sections add optional controls without changing the basic workflow for simple packages.

- Use clear package names that customers and technicians can recognize.
- Set download and upload speeds in Mbps.
- Choose Active only when the package should be assignable to customers.

Screenshot placeholder: package create form

3. Burst Package Configuration

Burst settings let a package temporarily exceed its normal speed. This is useful for responsive browsing and short downloads.

When Enable burst is No, burst fields are hidden and old burst values are not applied.

- Burst download speed and upload speed define temporary maximums.
- Burst threshold defines when the router allows burst.
- Burst time defines how long the burst can last.
- Priority and limit-at are available for queue planning.

```
Example: normal 10M/5M, burst 20M/10M, threshold 8M/4M, burst time 30s/30s
```

4. Scheduled Package Configuration

Schedule settings make a package available only during specific times. The UI includes the helper text: Enable to make this package available only during specific times.

When scheduling is disabled, schedule start, end, days, and recurring settings are cleared for that package.

- Use start and end time for time-based packages.
- Choose days of week for weekday, weekend, or night plans.
- Use recurring schedules for packages that repeat every week.

Screenshot placeholder: schedule controls

5. Package Visibility Rules

Available on all MikroTik controls whether the package can be used on every router in the ISP or only on selected devices.

Hide this package from the client controls whether customers should see the package in customer-facing flows.

- If Available on all MikroTik is Yes, selected router links are cleared.
- If No, choose one or more MikroTik devices.
- Hidden packages can still be used internally by staff if needed.

6. Add a MikroTik Device

The add flow is simplified. The user enters only essential onboarding details, while detailed device information can be filled after provisioning or later by a RouterOS sync worker.

The required first step is Connection / MikroTik Identity.

- Enter the MikroTik identity/device name.
- Enter host, API port, API username, and password if available.
- Choose whether the router supports PPPoE, Hotspot, or both.

Screenshot placeholder: Add MikroTik Device step 1

7. Provisioning Command Flow

After saving a router, StudyRoom Connect shows the provisioning command. The user copies it and pastes it into MikroTik Terminal.

The command fetches the public Laravel provisioning URL and imports the returned .rsc script.

- Click Copy command.
- Paste into MikroTik Terminal.
- Wait for command execution.
- The system marks the device provisioned when the public script is fetched.

```
/tool fetch mode=http url="http://localhost/billing-system/provision/TOKEN"
dst-path=studyroom_provision.rsc
:delay 2s
/import studyroom_provision.rsc
```

8. Automatic Device Information

The system now has fields ready for router details such as board name, RouterOS version, architecture, uptime, CPU usage, memory usage, last seen, and service status.

In this patch, these fields are stored and displayed. A future RouterOS API worker can populate them live from the router.

- Board name and identity identify the router.
- CPU and memory help technicians detect overload.
- Last seen confirms connectivity.
- Hotspot and PPPoE statuses show service readiness.

9. MikroTik Management Page

The MikroTik list now shows operational columns similar to the reference screenshots: board name, provisioning, CPU, memory, status, and remote Winbox.

The View page shows a full device profile and management actions.

- View opens full router details.
- Provisioning Command opens the onboarding command page.
- Enable Winbox generates access details.
- Action forms show success/error feedback through Laravel flash messages.

Screenshot placeholder: MikroTik list and detail page

10. Remote Winbox Access

Remote Winbox access creates a managed endpoint, username, and autogenerated password from the MikroTik page.

Passwords are generated in the SW-XXXX-XXXX-0000 format and stored through Laravel encrypted casts.

- Use Enable Winbox to create initial access.
- Use Regenerate Winbox to rotate the password.
- The provisioning script enables the Winbox service when credentials exist.
- Protect Winbox endpoints with VPN or firewall rules in production.

```
Example endpoint: router-host:8291
Example username: studyroom-winbox
Example password: SW-ABCD-WXYZ-1234
```

11. Reprovisioning a Router

Reprovisioning generates a new router provisioning token and returns the user to the command page.

Use reprovisioning when the router was reset, a command expired, or setup needs to be repeated.

- Confirm before reprovisioning.
- Copy the new command.
- Run it in MikroTik Terminal.
- Old commands should no longer be used after a new token is created.

12. Sync Hotspot Files and Router Time

The management page includes actions for Sync hotspot files and Sync Router Time.

At this stage these actions update local sync status safely. A RouterOS API worker can later process the queued sync states.

- Hotspot file sync prepares hotspot portal or asset synchronization.
- Router time sync prepares NTP/time correction operations.
- Both actions are scoped through the same ISP authorization checks.

13. Troubleshooting Common Errors

Most provisioning issues come from browser URL configuration, MikroTik terminal restrictions, wrong host credentials, or device mode limitations.

If the device reports device mode not allowed, run the device-mode command shown on the provisioning page, power-cycle the router, then retry.

- Check APP_URL is reachable by the router.
- Confirm Apache is running.
- Confirm the router can resolve the host.
- Check that Winbox and API ports are not blocked.
- Use fresh provisioning tokens after reprovisioning.

14. Security Notes and Best Practices

Public provisioning URLs should contain long random tokens and should never expose admin pages or database errors.

Remote Winbox should be protected carefully because it can grant administrative router access.

- Rotate Winbox passwords after staff changes.
- Use VPN endpoints instead of public router IPs when possible.
- Keep RouterOS updated.
- Log provisioning requests.
- Limit who can enable or regenerate Winbox access.

15. Operational Checklist

Use this checklist when setting up a new ISP location or router.

It keeps package configuration, router linking, Winbox access, and customer provisioning in a repeatable order.

- Create or confirm the ISP account.
- Create packages with speeds, burst, schedule, and visibility.
- Link the MikroTik device.
- Enable PPPoE and/or Hotspot support.
- Enable or regenerate Winbox if remote access is needed.
- Run the provisioning command.
- Verify status, last seen, and service cards.
- Create customer provisioning tokens when ready.